Problem Sheet 1

discussion: Tuesday, 13.10.2026

For the implementations, please use either MATLAB or Python!

1.1. Consider the approximation of the integral $\int_0^1 f(x)$ by the trapezoidal rule

$$\int_0^1 f(x)\,dx \approx T(h) := \sum_{i=0}^{N-1} \frac{h}{2}\left(f(x_i) + f(x_{i+1})\right), \qquad x_i = ih, \qquad h = \frac{1}{N}$$

as well as the box rule

$$\int_0^1 f(x)\,dx \approx R(h) := \sum_{i=0}^{N-1} hf(x_i), \qquad x_i = ih, \qquad h = \frac{1}{N}.$$

a) Write one program that, given N and a function f, returns $T(h)$ and one that returns $R(h)$ with $h = 1/N$.

b) For $f(x) = e^x$ and $N(i) = 2^{i+1}$, $i = 0, \dots, 9$, set $\mathtt{h}(i) = 1/N(i)$. For these 10 values of h, compute the actual errors $\mathtt{err}(i) = \int_0^1 e^x\,dx - T(h(i)) = e - 1 - T(h(i))$. Plot the (absolute value) of the true error in a double logarithmic plot. That is, in `python` notation with the vectors `h` and `err` (both of length 10)

`matplotlib.pyplot.loglog(h, abs(err))`

Does the true error behave like Ch^2?
Do the same thing for the box rule as well. What error behaviour do you see?

1.2. Again, consider the approximation of the integral $\int_0^1 f(x)$ by the trapezoidal rule $T(h)$ from exercise 1.1. The aim is to design an error estimator as it was done for the "box scheme" in the lecture.

a) For the error estimator, make the ansatz

$$\int_0^1 f(x)\,dx - T(h) \stackrel{!}{=} Ch^2$$

and compute C from (calculated) $T(h)$ and $T(h/2)$. The error estimator is then $E(h) = Ch^2$.

b) Modify your program from exercise 1.1(a) such that it also computes the error estimator $E(h)$. For the test-case in exercise 1.1(b) do also compute the ratios $\mathtt{r}(i) := \mathtt{err}(i)/E(h(i))$ of the true error to estimated error. Plot the ratio `r` in a semilogarithmic plot versus h. That is, in `python` notation with the vectors h and `r` (both of length 10)

`matplotlib.pyplot.semilogx(`h`, r)`

1.3. Let C, $\alpha > 0$ and consider the function $h \mapsto f(h) = Ch^\alpha$. Why is this function a straight line in a `loglog`-plot? What is its slope? Other popular plotting schemes are, `semilogx` and `semilogy`. Which one would you use to plot functions of the form $N \mapsto Ce^{-bN}$? How would you proceed if you suspect that a function $h \mapsto f(h)$ has the form $f(h) = Ce^{-b/h}$? What if you suspect $f(h) = Ce^{-b/h^2}$?

1.4. Let the data $(0, 0)$, $(1, 2)$, $(4, 8)$ be given.

a) Write down the (quadratic) interpolating polynomial p in the Lagrange form, i.e., $p(x) = \sum_{i=0}^2 f_i \ell_i(x)$.

b) The quadratic interpolating polynomial $p(x) = a_0 + a_1 x + a_2 x^2$ can also be determined by solving a linear system of equations for the coefficients a_0, a_1, a_2. Formulate the system and solve for the a_i.

c) Check your approximation $p(2)$ using the `numpy` routines `polyfit(x, f, n)` and `polyval`. Note that `polyfit` also returns the coefficients a_i of part b). Compare.
Note: The use of `polyfit` and `polyval` in the context of polynomial interpolation should rather be viewed as a "quick and dirty" method since `polyfit` is based on polynomial approximation in the monomial basis $\{1, x, x^2, \dots, x^n\}$ and therefore numerically unsuitable for large/largish n.

1.5. **a)** Write a routine that has as input the vectors $\mathbf{x}$ and $\mathbf{f}$ of knots and data (both vectors have the length $n+1$) and returns the value of the interpolating polynomial at the point 0, i.e., $p(0)$ and the polynomial p of degree n satisfies $p(\mathbf{x}_i) = \mathbf{f}_i$ for all i. To simplify matters, you may use `polyfit` and `polyval`.

b) Based on a), write a function whose input is a function handle f and integers m, $n \in \mathbb{N}_0$. The output is a matrix $N \in \mathbb{R}^{(m+1)\times(n+1)}$ with entries $N_{j,k} := p_{j,k}(0)$, where $j = 0, \ldots, m$, $k = 0, \ldots, n$ and $p_{j,k}$ is the polynomial of degree k that interpolates the data $(h_{j+\ell}, f(h_{j+\ell}))$, $\ell = 0, \ldots, k$, where $h_i := 2^{-i}$.

Remark: This matrix N is the so-called *Neville*-scheme for interpolation although it is usually set up differently. (This will be done later in class.)

c) Apply your routine of b) to the function

$$f(h) = \frac{e^h - 1}{h}$$

and select $m = 10$ and $n = 2$. You expect the columns of N to converge to $\lim_{h\to 0} f(h) = 1$. To check that, plot in a log-log plot h vs $|N(:,k) - 1|$ for all three columns $k = 1, 2, 3$. What convergence behavior do you observe for these 3 graphs?

1.6. Let $f(x) = \frac{x^2}{1+x^2}$. Use the knots $x_0 = 0$, $x_1 = 1$, $x_2 = 2$ and values $f_i = f(x_i)$.

a) Compute the linear interpolation polynomial in x_0 , x_3 and the cubic interpolation polynomial in all knots.

b) Use both interpolation polynomials to compute approximations to $\int_0^3 f(x)dx$ and calculate the errors.